

Rice Growth-Stage, Cropping Intensity & Production from S1+S2 Fusion

MOGPR Sentinel-1 + Sentinel-2 — Conclusions (Java, 2024-2026)

Dr. Firman Hadi

June 2026

Can we map rice growth stage — and from it cropping intensity and production — better than with Sentinel-1 alone?

- The TA-2026 reports built a strong **S1-only** paddy map (95% CV, ~3.0 M ha).
- But their **growth-stage** model was *not validated* (32% vs field data) and they parked **S1 + S2 fusion** as long-term future work.
- This work builds and validates that fusion.

- **MOGPR** fuses Sentinel-1 VH (SAR, all-weather) + Sentinel-2 NDVI (optical) into one gap-filled curve per pixel.
- MOGPR is a **feature generator**, not the final model.
- A trained **MiniROCKET** + **LightGBM** classifier reads the fused curve $\rightarrow 6 / 3$ growth phases.
- Production = area in generative phase $(4+5) \times 5.8$ t/ha, per 12-day period, Java-wide, 50 m.

Finding 1 — Fusion beats SAR-only for phase

Generative-phase F1 (leave-one-region-out):

Method	F1
Rule-based on SAR curve	~22%
VH-only	52%
VH + MOGPR optical	67-68%
idmai VH-CNN (in-sample)	64%

- **Optical is the dominant signal** for phase; SAR alone cannot carry it.
- This is the validated phase model the reports lacked.

Finding 2 — Multi-season training is non-negotiable

A model trained only on the wet season **collapses** on the dry season:

Training	Dry-season accuracy	Dry-season generative F1
Wet-only	36.9%	60.7%
Wet + dry (multi-season)	78.6%	91.6%

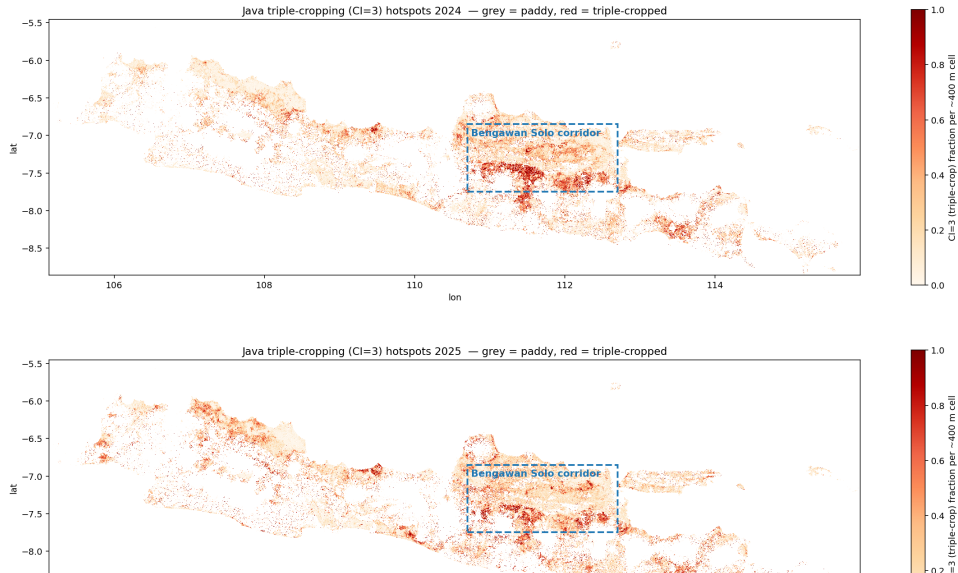
- Fixed using existing dry-season field labels (no new data needed).
- Independently reproduces the S1-only literature's central finding — on the fused signal.

Finding 3 — Production (method demonstration)

Year	Cropping intensity	Production
2024	2.18	28.8 Mt
2025	2.22	29.4 Mt
2024-2025 combined	2.22	~29.5 Mt/yr

- Harvest area ~5.0 M ha = **close to BPS luas panen (~5.34 M)** — an independent sanity check.
- Detected-paddy denominator (2.28 M ha), flat 5.8 t/ha.

Finding 4 — A real interannual signal



Java cropping intensity — combined 2024-2025

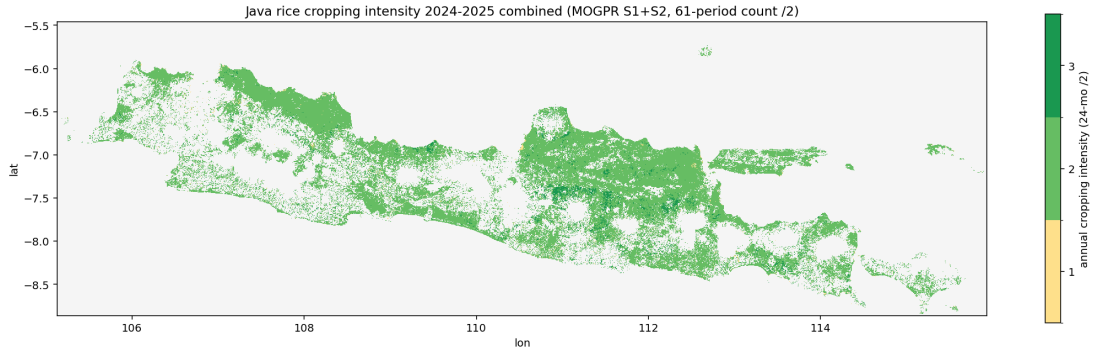


Figure 2: Annual cropping intensity, 24-month combined count

- **Method demonstration, not official numbers:** detected-mask denominator + flat yield.
- Accuracy is **optimistic** — validated on a random (not spatial) split, same-campaign labels.
- 6-class is weak (~51%); robust products are **3-class** and the **generative** class.

- **S1 + S2 MOGPR fusion materially improves rice growth-stage mapping over S1-only**, and turns it into a working **cropping-intensity and production** product for Java.
- It delivers — ahead of schedule — the fusion the program had parked as future work.
- The S1 binary-paddy results stand; this **adds** a validated phase -> IP -> production capability.

Next step — the decisive one

TA-2027 multi-DI field validation (Klambu, Rentang, Jatiluhur, Semarang, E. Java, rainfed):

- Spatial leave-one-DI-out test (not random) + a dry-season campaign.
- Validate IP against tracked actual harvests; production vs KSA/BPS.
- Converts “promising demonstration” -> “operationally trustworthy.”
- **Pilot already prepared:** 100 lag-adjusted Semarang points awaiting survey.